6. Area Under Curve (AUC) and ROC Curve

It is useful for binary classification tasks. The AUC value represents the probability that the model will rank a randomly chosen positive example higher than a randomly chosen negative example. AUC ranges from 0 to 1 with higher values showing better model performance.

1. True Positive Rate(TPR)

Also known as sensitivity or recall, the True Positive Rate measures how many actual positive instances were correctly identified by the model. It answers the question: "Out of all the actual positive cases, how many did the model correctly identify?"

Formula:

$$TPR = \frac{TP}{TP+FN}$$

Where:

- TP = True Positives (correctly predicted positive cases)
- FN = False Negatives (actual positive cases incorrectly predicted as negative)
-

2. True Negative Rate(TNR)

Also called specificity, the True Negative Rate measures how many actual negative instances were correctly identified by the model. It answers the question: "Out of all the actual negative cases, how many did the model correctly identify as negative?"

Formula:

$$TNR = \frac{TN}{TN + FP}$$

Where:

- TN = True Negatives (correctly predicted negative cases)
- FP = False Positives (actual negative cases incorrectly predicted as positive)

3. False Positive Rate(FPR)

It measures how many actual negative instances were incorrectly classified as positive. It's a key metric when the cost of false positives is high such as in fraud detection.

Formula:

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

Where:

- FP = False Positives (incorrectly predicted positive cases)
- TN = True Negatives (correctly predicted negative cases)

4. False Negative Rate(FNR)

It measures how many actual positive instances were incorrectly classified as negative. It answers: "Out of all the actual positive cases, how many were misclassified as negative?"

> Formula:
>
> $$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}}$$

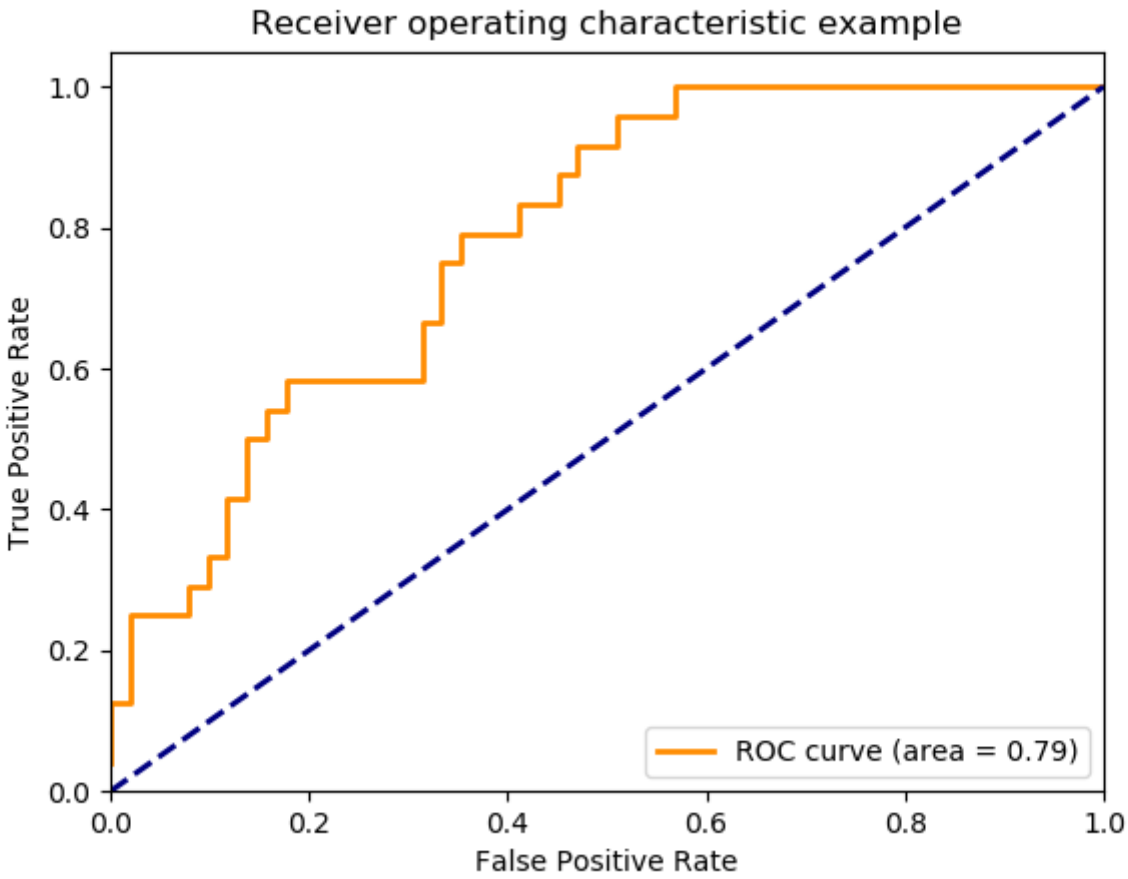
Where:

- FN = False Negatives (incorrectly predicted negative cases)
- TP = True Positives (correctly predicted positive cases)

ROC Curve

It is a graphical representation of the True Positive Rate (TPR) vs the False Positive Rate (FPR) at different classification thresholds. The curve helps us visualize the trade-offs between sensitivity (TPR) and specificity (1 - FPR) across various thresholds. Area Under Curve (AUC) quantifies the overall ability of the model to distinguish between positive and negative classes.

- AUC = 1: Perfect model (always correctly classifies positives and negatives).
- AUC = 0.5: Model performs no better than random guessing.
- AUC < 0.5: Model performs worse than random guessing (showing that the model is inverted).



ROC Curve for Evaluation of Classification Models